

# Théo Morales, Ph.D. in Computer Science

3D Computer Vision & Machine Learning

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in Théo Morales

🌐 <http://theomorales.com/>



## Employment history

- 2024 – 2026 ■ **Postdoctoral researcher.** School of Computer Science & Statistics, Trinity College Dublin, Ireland.  
Research: 4D reconstruction; dynamic Gaussian Splatting; 4D representations
- 2023 ■ **Short-term lecturer.** Fuzhou University & Maynooth University, China.
- 2020 – ... ■ **Programming Tutor.** online and in Dublin, Ireland.
- 2019 – 2020 ■ **Computer Vision engineer (R&D).** Dopli / Cisteme / XLIM, France.  
Topics: edge AI; object tracking; model training & deployment
- 2019 ■ **Research assistant.** Artificial Intelligence in Robotics group, Aarhus University, Denmark.  
Topics: ROS; Gazebo; sim2real; UAV robotics
- 2017 – 2019 ■ **Part-time software developer.** Trifork, Aarhus, Denmark.  
Topics: web development; iOS development
- 2016 – 2017 ■ **Part-time software developer.** Asimut Software, Aarhus, Denmark.  
Topics: full-stack development

## Education

- 2020 – 2024 ■ **Ph.D. Computer Science, Trinity College Dublin.**  
Thesis title: *Designing Hand-Object Interaction Representations for Better Grasp Priors.*  
Topics: meta-learning; generative models; diffusion models; 3D representations
- 2017 – 2019 ■ **M.Sc. Computer Engineering, Aarhus University.**  
Thesis title: *Synthetic Images for Convolutional Neural Networks in Autonomous Drone Racing.*  
Topics: CNNs; ROS; sim2real; OpenGL; robotics
- 2016 – 2017 ■ **B.Eng. Information and Communication Technology, VIA University College.**  
Thesis title: *Pizza Ordering Management System.*  
Topics: TDD; REST API
- 2013 – 2016 ■ **DUT. Information and Communication Technology, IUT La Rochelle / VIA University College.**  
Undergraduate IT diploma in Embedded Systems & Signal Processing.

## Publications

### Conference proceedings

- 1 T. Morales, O. Taheri, and G. Lacey, “A versatile and differentiable hand-object interaction representation,” in *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2025, pp. 23–33. [DOI: 10.1109/WACV61041.2025.00013](https://doi.org/10.1109/WACV61041.2025.00013).
- 2 S. Ravi, P. Climent-Perez, T. Morales, C. Huesca-Spaurani, K. Hashemifard, and F. Florez-Revuelta, “ODIN: An OmniDirectional INdoor dataset capturing Activities of Daily Living from multiple synchronized modalities,” in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, Los Alamitos, CA, USA: IEEE Computer Society, Jun. 2023, pp. 6488–6497. [DOI: 10.1109/CVPRW59228.2023.00690](https://doi.org/10.1109/CVPRW59228.2023.00690).

- 3 T. Morales and G. Lacey, “A new benchmark for group distribution shifts in hand grasp regression for object manipulation. can meta-learning raise the bar?” In *NeurIPS 2022 Workshop on Distribution Shifts: Connecting Methods and Applications*, 2022.  URL: <https://openreview.net/forum?id=IKbA3QS7c8X>.
- 4 T. Morales, A. Sarabakha, and E. Kayacan, “Image generation for efficient neural network training in autonomous drone racing,” in *2020 International Joint Conference on Neural Networks (IJCNN)*, 2020, pp. 1–8.  DOI: 10.1109/IJCNN48605.2020.9206943.

## Skills

|             |                                                                                                                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expertise   |  Mathematics for ML, Computer Vision, Computer Graphics, 4D reconstruction, 3D Gaussian Splatting, Diffusion Models, Transformers, ... |
| MLOps       |  Kubeflow, Docker, Kubernetes                                                                                                          |
| Languages   |  Native in French, Fluent in English, near working proficiency in Spanish.                                                             |
| Programming |  C, C++, Python, Java, Zig, CUDA, Bash, ...                                                                                            |
| Misc.       |  Academic research, teaching, training, consultation, $\LaTeX$ typesetting and publishing.                                             |

## Miscellaneous experience

### Achievements

2025  **Awarded funding**, Research funding for a project with Dolby.

### Conference reviewing

2024 – ...  **Reviewer**. CVPR, NeurIPS.

## References

Available on request.